CoE shell cosmology: methods and results, Oct 9–10, 2026

This note says what we did and how. Each section starts with a plain-language account, then gives the formula, and names the script that does the step. All paths are under `/workspace/clump_scan/` unless written in full. The results are the same as in the earlier results-only summary, which is kept as `notes_summary/coe_summary_2026-10-09_10_results_only_backup.md`. Numbers were checked against the source JSON files.

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0. Common model setup (used by every section)

0.1 The CMB residual we try to cancel

Plain words. The starting point is the shell model of the CMB anisotropy around the ring. It gives our velocity and gravitational redshift relative to the Centre of Expansion (CoE). We divide the observed dipole-like curve (the Migkas best-fit curve, 360 points at 1° steps around the ring) by the model and look at what is left. That leftover is the target the bodies have to cancel.

Formula (`cmb-cancel.py`, `scan.py`). Let $G(\ell)$ be the Migkas curve normalised to its mean, and $S(\ell)$ the shell-model prediction (motion Doppler factor times gravitational factor). Then

$$S = \frac{\gamma_o(1-\boldsymbol{\beta}_o\cdot\widehat{n})}{\gamma_e(1-\boldsymbol{\beta}_e\cdot\widehat{n})}\sqrt{\frac{1-2\mu/r_e}{1-2\mu/r_o}}, \qquad \mathcal{R}(\ell) = \frac{G\,S}{\langle G\,S\rangle} - 1, \quad \mathcal{R} \leftarrow \mathcal{R} - \langle\mathcal{R}\rangle.$$

The data file is `/workspace/cosmology-course/notes/migkas-best-fit-curve-0-360-full.csv`. The best-fit parameters are in `cmb-cancel-story.json`.

0.2 Units: the "×" scale and the ±1× band

Plain words. Every leftover is quoted as its rms divided by the CMB level, 10^{-5} (the "floor"). So 1× means the leftover is as small as the real CMB fluctuations. On a plot, the ±1× band is the pale mint strip between -10^{-5} and $+10^{-5}$.

$$L = \frac{\mathrm{rms}(\mathcal{R}-w)}{\mathrm{FLOOR}}, \qquad \mathrm{FLOOR} = 10^{-5}.$$

Here w is the pattern made by all bodies (§0.3). With no bodies, $L_0 = \mathrm{rms}(\mathcal{R})/10^{-5} = 237.3\times$. The four-clump compensated baseline is 88.8×.

0.3 How a body contributes (the tidal kernel)

Plain words. Each void or cluster is treated as a point mass. Voids get a negative mass equal to the mass they are missing. A body pulls on the light coming in along each

sightline of the ring. What matters is the difference between its pull at the ring point (on the 14.3 Gpc sphere) and its pull on us, taken along the sightline. That difference is then calibrated into $\Delta T/T$ units with one constant. The mean over the ring is removed, because a constant offset is not observable.

Formula (`shell_ensemble/ensemble.py: field3`; fast numba copy in `local_web/web_core.py: pattern`; vectorised as `sequential/seq_core.py: unit_pattern_fast`). Positions are in units of $R_{\rm vis} = 14.3$ Gpc. The ring points are $\widehat{n}_t = (\cos\theta_t, \sin\theta_t, 0)$, $t = 0..359$. For a body at $\boldsymbol{X}$ with mass M:

$$w_t = \frac{M}{K_G}\left[\frac{(\boldsymbol{X}-\widehat{n}_t)\cdot\widehat{n}_t}{|\boldsymbol{X}-\widehat{n}_t|^3} - \frac{\boldsymbol{X}\cdot\widehat{n}_t}{|\boldsymbol{X}|^3}\right] - \left\langle\ \cdot\ \right\rangle_t, \qquad K_G = \frac{7.7\times10^{51}}{0.0039}\ \text{kg}.$$

The web pattern is the sum over bodies, $w = \sum_i w^{(i)}$. It is linear in mass, which makes all fits linear in M. Densities follow the project's "exp-lock" law,

$$\rho(\boldsymbol{x}) = \rho_0\, e^{-(|\boldsymbol{x}-\boldsymbol{C}|-D)/L}, \qquad \rho_0 = 2.9\times10^{-27}\ \text{kg m}^{-3},\ D = 58.2\ \text{Gpc},\ L = 922.65\ \text{Gpc},$$

with C the CoE position (58.2 Gpc away). Across the 30 Gpc region the density changes by only ±3%.

0.4 Ring modes: m = 1–12 vs m ≥ 13, and the symmetric share

Plain words. The leftover around the ring is split into Fourier modes. Modes 1–12 are large scales, which a modest number of distant bodies can always cancel. Modes 13 and up are fine-scale ripple, which mostly comes from bodies close to the ring. The "symmetric share" measures how much of the leftover is mirror-symmetric about the CoE axis. The real CMB anomalies are mostly asymmetric.

Formula (`sequential_open/fit_on_web.py: spec`). Angles are measured from the CoE axis, $\phi = \theta - \theta_{\rm CoE}$:

$$a_m = \tfrac{2}{N}\sum_t v_t\cos(m\phi_t), \quad b_m = \tfrac{2}{N}\sum_t v_t\sin(m\phi_t), \qquad m = 1..12.$$

- **m1–12:** $\sqrt{\text{rms}_{1\text{–}5}^2 + \text{rms}_{6\text{–}12}^2}$ of the reconstruction from a_m, b_m.
- **m13+:** the rms of what remains after subtracting that reconstruction.
- **Symmetric share:** $\text{sym} = \frac{\sum_m a_m^2}{\sum_m (a_m^2 + b_m^2)}$. The cosine terms are the part that is even about the axis. Pure noise gives about 50%.

0.5 Modelled region and placement rules

- **Region** (`sequential_open/open_region.py`, mode `tidalcore`). Our 14.3 Gpc sphere plus the shell out to 30 Gpc from us, including the part of the CoE core inside 30 Gpc. 30 Gpc is the effective tidal depth.
- **No-straddle rule.** No sphere may cross the CMB ring sphere. It is either wholly inside ($|\boldsymbol{x}|+r \le 14.2$) or wholly outside ($|\boldsymbol{x}|-r \ge 14.4$). In the edit fitters this is written as the surface staying at least 0.1 Gpc off the ring circle: $d_{\rm ring}(\boldsymbol{x}) - r \ge 0.1$, with $d_{\rm ring} = \sqrt{\left(\sqrt{x^2+y^2} - 14.3\right)^2 + z^2}$.

- **Other rules.** Nothing within 0.5 Gpc of us ($|\boldsymbol{x}|-r \geq 0.5$). Whole sphere inside 30 Gpc. No overlaps: $|\boldsymbol{x}_i - \boldsymbol{x}_j| \geq r_i + r_j + 0.02$ Gpc.

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1. The brief (LaTeX)

What was done

- Cleared up the 0.58× vs 0.34× mismatch. 0.58× is from the brief (11 objects); 0.34× is from Writer's in-plane sketch (12 objects). The brief's runs are the source of record; Writer's tiled shell stays as an illustration.
- Fig 2 reworked. Panel (a) is the leftover after each step. Panel (b) is now a plain step-by-step demo of placing masses in the side view.
- Pages 8–10: all nine examples sorted globally from largest to smallest leftover (numbered 1–9).
- Added the patch-ratio result, the asymmetry test, a light-through-structure section (Sachs–Wolfe, ISW, Rees–Sciama, lensing), and the faster-than-light recession note (caveats plus one clause in the conclusions).
- Layout: page 13 charts stacked vertically, legends below graphs, taller charts, references not forced onto a new page, arXiv links.

How. The leftover-per-step curves are the joint-fit outputs (§3). The asymmetry test is the symmetric share of §0.4.

2. Figure standards adopted

- Every residual-around-the-ring plot shades the ±1× CMB band (pale mint, §0.2).
- Side views show only the modelled region (§0.5), sliced at $|z| < 1.5$ Gpc, with bodies drawn to scale.
- The CoE direction is drawn as a vector from us toward the CoE (`sequential_open/common.py: frame30`).
- Before/after views sit side by side with no arrows. Clusters are red, voids blue, and anything that moved is green in both views. The residual (before and after) goes underneath (`edit_before_after.py: residual_below`).
- One run per image, moderate fonts, legends below. Don't cram.
- Area-change maps use Doppler colours: blue = compressed, red = expanded, white = no change (diverging RdBu with a symmetric log scale, `edit_before_after.py: change_map`).
- Exception: on force maps each body is drawn as a body-sized arrow along its net force (`net_force_arrows.py`).

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3. Placement fits and solution space

3.1 The joint fitter

Plain words. We choose a set of bodies and fit all of their positions and masses together, so that their combined pattern cancels the residual. The fit is bounded least squares. The placement rules enter as penalty terms that are switched on only when a rule is broken. We start the fit from several points and keep the best start that obeys every rule.

Formula (joint fits `joint_add.py`, `fit_on_web.py: fit_one`; batch edit fits `minedit_ent.py: fitset`). For the parameters q of the selected bodies,

$$\min_{q} \left\| \frac{T - \sum_{i \in \text{batch}} M_i(q)\, K(\boldsymbol{x}_i(q))}{\text{FLOOR}\sqrt{360}} \right\|^2 + \lambda^2 \sum_{k} \left[\max(0, g_k(q)) \right]^2.$$

- T is what the batch has to supply: the residual minus the pattern of everything else.
- g_k are the rule violations (region, ring clearance, overlaps).
- λ starts at 30 and is multiplied by 10, up to 4 times, until every violation is at most 10^{-3}.
- Solver: `scipy.optimize.least_squares`, method `trf`, `x_scale="jac"`, Jacobian by finite differences.

3.2 Growth runs (building webs from empty)

Plain words. The webs are grown one object at a time. Each new object's placement depends on what is already there, and overlaps are strictly forbidden.

How (`sequential/scenario_b.py`, run with the open region by `sequential_open/run_b_open.py`). Voids have $R = 1.5$ Gpc and $\delta = -0.70$. Clusters have $\delta = +100$. Each void's deficit is paid back by 6 clusters. The growth rules:

- **random:** every object is dropped at a random free spot. This is the unstructured control.
- **foam_rim:** a new void grows next to an existing void; its 6 clusters then go one by one onto rim spots that touch other voids (the junctions).
- **cmb_aware:** like foam_rim, but each placement is the best of about 30 allowed spots for the leftover.

The variants filled the core, the visible sphere, or both (`make_tidalcore.py`).

3.3 Results

- Joint inside-plus-outside fits were run with the core open and bodies allowed inside our sphere.
- Growth runs: random and foam rule, with the core filled, the visible sphere filled, or both.
- Statistical framing: draw random universes from real statistics, estimate the rare tail, and measure the "good" region. Entropy is the surface to optimize.
- Symmetry: real CMB anomalies are mostly asymmetric, so we measure the symmetric share of every leftover.
- Large scales (m = 1–12) can always be cancelled. The web leaves fine-scale (m13+) texture.

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4. Void–cluster network rules

4.1 Generator and link census

Plain words. This is a “nothing special” random web built the way the real cosmic web is pictured (Icke & van de Weygaert 1987; van de Weygaert & Icke 1989). Voids are the cells of a random foam. Clusters sit at the corners where 4 or more cells meet, and their masses follow the measured cluster mass function. Links join each cluster to the voids around it.

How (`void_mass_network/vmn_gen.py`, rules and sources in `void_mass_network/rules.md`).

- **Void seeds:** Poisson points with a soft hard-core of 0.7× the mean spacing. Cells are Voronoi cells, with volumes found by Monte Carlo.
- **Void cores:** the void core is 77% of the cell volume at $1+\delta = 0.2$ (Cautun et al. 2014, arXiv:1401.7866).
- **Nodes:** nodes are the Delaunay circumcentres. Nodes closer than 0.15× the spacing are merged, which calibrates about 4.5 voids per cluster (Codis, Pogosyan & Pichon 2018, arXiv:1803.11477).
- **Clusters:** the top nodes by empty-sphere radius (with lognormal scatter) become clusters. Their masses are abundance-matched to Tinker 2008: the k-th heaviest has $n(>M)V = k - \frac{1}{2}$.

4.2 No-crossing rule

Plain words. Links are treated as thin tubes. Candidate links are tried shortest first. A link is dropped if it would cross a link already kept, or if it passes through another body.

How (`void_mass_network/planar.py`, `census.py`). The tube radius is $r_t = 24$ Mpc, which is 0.05× the median void radius. Two links that share no end node must stay more than $2r_t$ apart.

4.3 Results

- Node-link census: cluster–cluster, cluster–void and void–void links; connections per node, link lengths, angles between links; no-crossing rule.
- Voids effectively “attract”: they push matter out, drain the walls and merge.
- Literature: Clampitt & Jain (arXiv:1404.1834), whose real voids are about half as deep as ours.

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5. Periodic structure and blast-wave energy

5.1 Einasto checkerboard backgrounds

Plain words. Einasto et al. found a roughly regular grid of superclusters and voids with a repeat of about 120 h^{-1} Mpc. With $h = 0.7$ that is 171 Mpc. We laid such a grid through the visible sphere and computed its pull on the ring with the same kernel.

How (`checkerboard/cb_core.py`, `run_cb.py`, `main_fit.py`; summary in `checkerboard/summary.md`). Sources: Einasto et al. 1997, Paper I, astro-ph/9701018; Paper II Fig 2, astro-ph/9704127.

- **chess:** cubes of 85.7 Mpc (half a period) at alternating $\pm A$.
- **Einasto cell:** 171 Mpc cells with a void at each centre:

$$\delta = -A\,\frac{\cos kx + \cos ky + \cos kz}{\sqrt{1.5}}, \qquad k = \frac{2\pi}{171\ \mathrm{Mpc}}, \quad A = 0.10\ (\mathrm{rms}).$$

- **Amplitude:** $A = 0.10 \approx \sqrt{0.3} \times 0.19$. That is the regular share of the linear-CDM rms at the cell scale (σ8 = 0.81).
- **Summation:** only cells within 1 Gpc of the ring (and 1 Gpc around us) are summed cell by cell. The far field of a periodic pattern cancels.

5.2 Radial (blast-wave) shells and the quasi-lattice

How (`sequential_open/wave_bg.py`, `fig_wave_lattice.py`, `radial_inside/ri_core.py`).

- **Radial wave:** concentric shells about the CoE, $\delta = A\cos(2\pi r_{\mathrm{CoE}}/\lambda + \varphi)$, with $\lambda = 150$ or 171 Mpc.
- **Quasi-lattice:** $\delta = \frac{A}{3}(\cos kx' + \cos ky' + \cos kz')$ on jittered coordinates x'.
- **How the pull is computed:** the coherent spherical part is exact (shell theorem, analytic $\delta M_{\mathrm{enc}}(r)$). The jittered part within 1 Gpc of the ring is summed over 37.5 Mpc cells with Plummer softening of one cell.

5.3 Energy estimates (order of magnitude)

Plain words. Two rough checks of how much energy such a wave carries. The first is the kinetic energy of the matter moving in a density wave. The second is the energy of a point explosion big enough to drive a shell out to that radius in the available time (Sedov–Taylor).

Formulas. For a sound-like density wave of contrast δ and wave speed v_s in a region of mass $M = \rho V$:

$$E_{\mathrm{wave}} \sim \tfrac{1}{2}\,\delta^2 \rho\, v_s^2\, V = \tfrac{1}{2}\,\delta^2 \left(\frac{v_s}{c}\right)^2 Mc^2.$$

Sedov–Taylor blast wave: $R(t) = \xi\,(Et^2/\rho)^{1/5}$ with $\xi \approx 1.15$, so

$$E_{\mathrm{ST}} \approx \rho R^5/(\xi^5 t^2).$$

These were evaluated inline in the session; no script was saved.

Results

- Einasto et al. 1997 periodicity, ~120 h^{-1} Mpc ≈ 171 Mpc: checkerboard and radial-shell backgrounds across the visible sphere.
- Wave energy (order of magnitude): ~10^{63}–10^{65} J in our sphere, 10^{65}–10^{67} J in the shell, ~10^{74} J total from Sedov–Taylor. This implies a faster-than-light edge, flagged as next-stage relativistic LTB work.

- The model ignores dark matter and compares observables only.

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6. Minimal-edit tests (start from a random filled web, edit to reach CMB level)

6.1 Method

Plain words. Take a finished web and change as few bodies as possible to bring the leftover down to the CMB level. Each round, we pick the bodies that promise the biggest gain if moved or resized. We fit them together (§3.1) and keep the round only if the leftover goes down.

Parameters per edited body (`sequential_open/minedit2.py`, `minedit_ent.py`): (x, y, z, a).

- a scales the mass. For a void, the deficit is a times the original, re-weighted by the local density; the radius is $R = R_0 a^{1/3}$.
- For a cluster, $M = aM_0$ and $r = r_0 a^{1/3}$.
- $a \in [a_{\min}, 10]$, where $a_{\min}$ keeps void radius ≥ 0.5 Gpc and cluster mass ≥ 0.1 M_0. With deletion allowed, $a \to 0$ means deleted.

Leverage (which bodies to pick). Linearise each body's pattern in its 4 parameters to get a Jacobian J (4×360). The best step and its gain are

$$\Delta = (JJ^\top + \mu I)^{-1} J r, \qquad \text{gain} = 2\Delta \cdot (Jr) - \Delta^\top (JJ^\top)\Delta,$$

with the move capped at 5 Gpc and r the current residual. With deletion allowed, the exact deletion gain $\| r \|^2 - \| r + C_i \|^2$ is also considered.

Batches and starts

- Batch size: $\text{clip}(n_{\text{edited}}/6,\ 2,\ 24)$.
- Starts: as found; 2 random nudges within 5 Gpc; one random spot within 15 Gpc.
- Every 5 rounds, the most recent ≤ 60 edited bodies are refitted jointly.

Stop rules. Stop at ≤ 1×, or at KMAX edits. The plateau rule stops a run when less than 1% gain came over the last 300 edits.

Fine-scale pass (FINE=1). The linear leverage cannot see that a near-ring body should move several Gpc. So near-ring bodies (within 6 Gpc) are ranked by their own m13+ power and fitted in batches of 12. Their starts are in place, pushed 2–8 Gpc away from the ring, and random. Each accepted batch is followed by a 24-body leverage "absorb" refit.

6.2 The ring rules

- **First attempt:** the fitter piled masses onto the CMB ring, because a mass right at the ring has huge leverage on the fine-scale modes.
- **3 Gpc backstop:** we added a hard rule that every body surface stays at least 3 Gpc from the ring. Bodies grown inside the band had to move out, shrink, or be deleted.

This was a judgment call, not physics, and it caused artificial pile-up just outside 3 Gpc.
- **Entropy rule (replacement):** no edit may make the web more ordered than it started. Details in §6.3.

6.3 The entropy rule: 7 components and their binning

Plain words. We measure how "spread out" the web is in seven ways. An edit is allowed only if none of the seven goes down compared with the starting web. That rules out forming a ring, a pile-up, a hole or a regular pattern.

Formulas (`sequential_open/entropy_core.py`). All bodies alive inside 30 Gpc are used.

- **Volume-weighted (position) entropies,** with w_b the share of region volume in bin b (Monte Carlo, 4×10^6 points):

$$H = -\sum_b p_b \ln \frac{p_b}{w_b} \leq 0,$$

  which is 0 when the bodies are uniform in volume.
  1. **radial:** 30 equal-volume shells around us, 0–30 Gpc.
  2. **ring:** distance from the CMB ring circle. There are 24 bins of equal tube volume out to 14.3 Gpc, with bin index $24\,(d/14.3)^2$; the first bin reaches 2.9 Gpc. One more bin holds everything else.
  3. **angular:** 12 × 24 equal-area sky cells ($\cos\vartheta \times \varphi$).
- **Shannon entropies of histograms,** $S = -\sum p \ln p$, on fixed bins taken from the start (40 bins from 0 to 1.5× the 99.5th percentile):
  4. **link_cc:** nearest cluster→cluster distance.
  5. **link_cv:** nearest cluster→void distance.
  6. **link_vv:** nearest void→void distance.
  7. **ncount:** number of neighbours within 3 Gpc (integer histogram).

How it is enforced. As a hard check after every fit ($S_k \geq S_{k,0}$ for all k). Inside the fit there is also a one-sided soft penalty for components 1–3, using linear-interpolated (differentiable) bin counts, with weight 3000 and margin 2×10^{-5}.

6.4 Results

- **First fitter:** piled masses on the CMB ring. We reverted to the joint fitter with a 3 Gpc ring backstop and filled the core cap.
- **With deletion allowed:** reached 0.87–1.00×, but only by deleting bodies near the ring.
- **No deletion (move and resize only), with the 3 Gpc backstop** (`minedit2/*_nodel_reloc.json`): two foam webs reached 1×.
  - Foam s3 needed 489 edits (2.8% of the web) to reach 0.995×.
  - Foam s1 needed 614 edits (3.6%) to reach 0.954×.
  - Near-edge bodies moved a median of ~2 Gpc and others under 1 Gpc. Most masses changed by less than 2×.
- **Under the entropy rule** (`minedit_ent/*_entropy.json`, table in `minedit_ent/table.md`) none of the six grown webs reaches 1×.

- Best is foam s2: 1223× → 2.40× with 638 edits (3.8%). m1–12 drops to ~0.01×; the rest is m13+ ripple.
- No pile-up at the ring. Bodies whose surface is within 3 Gpc of the ring went 221 → 230, so the edits partly refill the gap left by the growth rule.

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7. Mesh, area, entropy and energy (foam s2)

7.1 Non-crossing planar mesh and its faces

Plain words. Take the bodies in the side-view slab, join near neighbours with straight links that never cross, then measure the polygons (faces) the links enclose.

How

- **Links** (`sequential_open/mesh_view.py`): candidates are each slab node's 10 nearest slab neighbours in 3D, within 6 Gpc. They are tried shortest first. A link is dropped if it passes through another body (the void slab cross-section, 0.95 R) or comes within $2r_t$ of a kept link that shares no end node. This is the `planar.py` rule applied in the in-plane projection.
- **Faces** (`sequential_open/mesh_faces.py`): a half-edge walk of the planar graph. The outer face of each component (negative signed area) is dropped, as are faces cut by the 30 Gpc edge. Faces above a size threshold are treated as slab holes.
- **Area** by the shoelace formula: $A = \frac{1}{2}\left|\sum_k (x_k y_{k+1} - x_{k+1} y_k)\right|$.

7.2 Full Delaunay triangulation

Plain words. Joining every node with non-crossing links until no more fit gives triangles only. That is the Delaunay triangulation. It leaves no gaps: every spot inside the outermost nodes lies in some triangle.

How (`sequential_open/tri_view.py: tri`; coloured map `tri_coloured.py`; histograms `tri_hist.py`; degree, length and field view `tri_conn_field.py`). We use `scipy.spatial.Delaunay` on the in-plane projection of the slab nodes ($|z| < 1.5$ Gpc). The final runs use no cutoff. An earlier long-edge cutoff left grey gaps and was removed.

Area-change map. Each AFTER triangle is coloured by

$$\Delta A = A_{\text{after}} - A_{\text{before}}(\text{local}),$$

where $A_{\text{before}}(\text{local})$ is the area of the BEFORE triangle that contains the AFTER triangle's centroid (found with `Delaunay.find_simplex`). Colours: blue < 0 (compressed), red > 0 (expanded), white ≈ 0, on a symmetric log scale (linthresh 0.05 Gpc²).

Histograms. Triangle areas (log bins) for the whole slab and for triangles whose centroid lies 0–3 Gpc from the ring; connections per node (the degree from the edge list); link lengths. BEFORE is drawn grey, AFTER dark blue, stacked vertically.

7.3 Entropy and toy energy formulas

Plain words. Entropy here measures how evenly the area is shared among the cells. Energy is the gravitational energy along the links, plus a “surface” (wall) term for the length of the walls. A toy free energy $F = E - T_{\rm eff} S$ asks whether entropy could compete with gravity at all.

Formulas (`sequential_open/mesh_energy2.py`, outputs `mesh_energy.json` and `mesh_energy_compare.json`).

- **Area entropy and evenness:**

$$S = -\sum_i p_i \ln p_i, \quad p_i = \frac{A_i}{\sum_j A_j}; \qquad \text{evenness} = \frac{S}{\ln N} \in (0, 1] \ \ (1 = \text{all cells equal}).$$

- **Gravitational energy over the mesh links,** using the model’s signed masses (clusters $+M$, voids the negative deficit):

$$U = -G \sum_{\langle ij \rangle} \frac{m_i m_j}{r_{ij}}.$$

  Here $m_i m_j > 0$ (cluster–cluster or void–void) is bound and attracts, while $m_i m_j < 0$ (cluster–void) is unbound and repels.

- **Wall energy:** $E_{\rm wall} = \gamma\, v^2\, L_{\rm perimeter}$. The line density γ (kg/m) is set from the BEFORE cluster–cluster links: each cluster’s mass is shared over its links. Two speeds are used: $v = 300$ km/s, and $v = c$ as an upper bound.

- **Toy free energy:** $\Delta F = \Delta U + \Delta E_{\rm wall} - k_B T_{\rm eff}\, \Delta S$, with $k_B T_{\rm eff} = \langle \frac{1}{2} M v^2 \rangle$ over the slab clusters.

7.4 Results

- **Non-crossing mesh:** cells near the ring grow (median 0.60 → 1.20 Gpc² within 3 Gpc).
- **Full Delaunay, near the ring:** most spots are compressed, because the big gap triangles get split. But the population evens out: median 0.79 → 1.35 Gpc², area entropy 4.70 → 5.03.
- **Full Delaunay, whole slab:** median 0.59 → 0.66 Gpc², about 6 connections per node, median link 1.77 → 1.87 Gpc.
- **Energy:** near the ring it rises (~+23%); farther out it falls (~−14% at 3–6 Gpc). Overall the web is ~3% more bound. The entropy term is negligible next to gravity.
- **Density-field view** (`tri_conn_field.py`). Clusters are drawn as 3D Gaussians of width equal to the body radius; voids are top-hats of $\delta = -0.70$ with a tanh edge of width 0.1 R. On this view, overall clumpiness rises only 2–5%, and the void-trough area near the ring falls from 19% to 12%.

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8. Crystal analogy

8.1 FCC / fluorite reference

Plain words. In this crystal analogy, clusters sit on a face-centred-cubic lattice and voids sit in the tetrahedral holes. Each void is then surrounded by 4 clusters, a pyramid. This is the fluorite (CaF_2) arrangement. Octahedral holes (6 clusters around a void) are used for comparison.

How (`sequential_open/fcc_compare.py`, outputs in `sequential_open/fcc_lattice/`).

- Cubic cell edge $a = 171$ Mpc (the Einasto period). Masses: the web's median cluster mass and median void deficit.
- 3D Delaunay tetrahedralisation of the lattice box and of the whole foam s2 web. Degenerate co-spherical slivers (volume below 10^{-6} of the median) are removed after a 10^{-7} jitter.
- **Statistics:** tetrahedron volumes, node degree, link lengths.

  $$S_{\rm vol} = -\sum p_i \ln p_i, \quad p_i = \frac{V_i}{\sum V}; \qquad \text{evenness} = \frac{S_{\rm vol}}{\ln N}.$$

  A second measure is the Shannon entropy of the histogram of $\log_{10}(V/\bar{V})$ in 0.1-dex bins (the spread of cell sizes).
- **Link energy per node** in units of $GM_c^2/\bar{d}$, with $\bar{d}$ the mean spacing, so that different structures can be compared.

8.2 Imperfect-crystal ladder

How (`sequential_open/fcc_ladder.py`, figures `fcc_figs.py`, `ladder.json`). The ladder runs from perfect to random:

- **Perfect** FCC.
- **Thermal jitter:** Gaussian displacements with rms 5%, 15% and 30% of the spacing, $\sigma = f\,d/\sqrt{3}$ per axis.
- **Point defects** at 3% and 10% each: vacancies (clusters removed), interstitials (extra clusters at random spots), and voids moved from tetrahedral to octahedral holes.
- **Polycrystal:** 10 Voronoi grains with random orientations, spacing ±5%, 5% jitter. The grain boundaries are filled with Poisson points at the crystal density (boundary zone 0.6 a).
- **Fully random:** Poisson points.

Two compositions were run: "fluorite" (voids in all tetrahedral holes) and "web ratio" (1 void per 6 clusters). Each structure is plotted as a point on an entropy (x) vs link-energy-per-node (y) chart.

8.3 2D triangular crystal: Kawasaki anneal with Ewald sums

Plain words. A flat triangular lattice holds equal numbers of clusters and voids. We swap pairs of unlike sites at random and keep or reject each swap by the Metropolis rule while slowly cooling, so the total count of each kind never changes. That finds the lowest-energy arrangement. Long-range pulls in a periodic box are summed with the Ewald method.

Formulas (`sequential_open/tri2d/tri2d_core.py`, `tri2d_run.py`, `tri2d_figs.py`).

- **Energy:** $U = \frac{1}{2}\sum_{i \neq j} m_i m_j \, \Phi(\boldsymbol{r}_{ij})$, in units of GM_c^2/a. The sign convention $U = -G\sum m_i m_j / r$ makes like–like pairs bound.
- **2D-periodic Ewald potential of** $1/r$ with splitting parameter $\alpha = 6/\min(W_x, W_y)$:

$$\Phi(\boldsymbol{r}) = \sum_{\boldsymbol{n}} \frac{\operatorname{erfc}(\alpha|\boldsymbol{r}+\boldsymbol{n}|)}{|\boldsymbol{r}+\boldsymbol{n}|} + \frac{2\pi}{A}\sum_{\boldsymbol{k}\neq 0} \frac{\operatorname{erfc}(k/2\alpha)}{k}\cos(\boldsymbol{k}\cdot\boldsymbol{r}) - \frac{2\sqrt{\pi}}{A\alpha},$$

  with the self-term $-2\alpha/\sqrt{\pi}$ on the diagonal. The full $N \times N$ matrix is built from translation invariance.
- **Kawasaki swap** of sites i and j (with $m_i \neq m_j$), using the field $h_i = \sum_k P_{ik} m_k$:

$$\Delta U = \delta_i h_i + \delta_j h_j + \tfrac{1}{2}\left(\delta_i^2 P_{ii} + \delta_j^2 P_{jj}\right) + \delta_i\delta_j P_{ij}, \qquad \delta_i = m_j - m_i,\ \delta_j = -\delta_i.$$

  For nearest-neighbour-only runs, the shared-neighbour correction for adjacent swapped sites is included.
- **Cooling:** geometric. Nearest-neighbour runs: T from 3 to 0.01 in 60 steps, $40N$ swaps. All-pairs runs: T from 5 to 0.01 in 40 steps, $15N$ swaps. Box 44 × 46 sites (about 2,000).

8.4 Compensated placement (2D and 3D)

Plain words. A void's missing mass has to go somewhere, so it is put into clusters in a wall around the void. Every void plus its wall then has zero net mass.

Formulas (2D: `sequential_open/tri2d/comp_place.py`; 3D: `sequential_open/compweb_gen.py`).

- **Void deficit and cluster mass:**

$$M_{\mathrm{def}} = |\delta_v|\,\rho\,\tfrac{4}{3}\pi R^3, \qquad M_c = \delta_c\,\rho\,\tfrac{4}{3}\pi r_c^3 \ \text{(excess mass)}.$$

- **Filling the wall:** clusters are added until $\sum M_c = M_{\mathrm{def}}$, with the last one trimmed. They go at distance R to $1.3R$ from the void centre (the compensating ridge of Hamaus et al. 2014, arXiv:1403.5499, and Clampitt & Jain 2015).
- **Cluster count:** this gives $N \approx N_0 + M_{\mathrm{def}}/\bar{M}_{c,\,\mathrm{eff}}$. A linear fit gives $N = 0.92 + M_{\mathrm{def}}/2.10\times10^{50}$ kg in 2D, and $N = 0.49 + M_{\mathrm{def}}/1.48\times10^{50}$ kg in the 3D web of §10.
- **2D ground state:** a Metropolis anneal of all-pairs U (minimum image). Moves are either a cluster anywhere in its own void's wall, or a void translated together with its clusters.

8.5 Results

- **FCC reference:** clusters on an FCC lattice with voids in the tetrahedral holes (4-cluster pyramids), 171 Mpc cell; octahedral holes for comparison.
- **Ladder from perfect to random:** defects, jitter, polycrystal. Our foam web sits *past* random in disorder and is the least bound. The edits nudge it slightly toward the lattice.

- **2D triangular crystal,** equal numbers, voids as bare negative mass: the lowest-energy state is full segregation. That is unphysical, because it ignores where the void's missing matter goes. With all-pairs energy the segregated state has no finite per-node limit (it keeps getting more bound with box size), i.e. gravitational collapse (`tri2d_size_scaling.json`).
- **Balance (compensation) rule:** each void's missing mass $M_{\rm def} = |\delta|\rho\frac{4}{3}\pi R^3$ sits in clusters around it, in a wall at R to $1.3R$, with $N \approx 0.9 + M_{\rm def}/2.1\times10^{50}$ kg (2D). At lowest energy the clusters bunch into knots where 2–3 voids meet, like the cosmic web (`tri2d/comp_results.json`).

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9. Forces

9.1 Pair-force calculation

Plain words. For every pair of neighbouring bodies we compute their mutual energy and force with the model's signed masses. Pairs closer than touching are softened, so overlapping bodies don't blow up.

Formulas (`sequential_open/pair_energy.py`, output `pair_energy.json`; map `net_force_arrows.py`).

$$U_{ij} = -\frac{G\,m_i m_j}{r_{\rm eff}}, \qquad \boldsymbol{F}_i = \sum_j \frac{G\,m_i m_j\,(\boldsymbol{x}_j - \boldsymbol{x}_i)}{|\boldsymbol{x}_j - \boldsymbol{x}_i|\,r_{\rm eff}^2}, \qquad r_{\rm eff} = \max(r_{ij}, R_i + R_j).$$

- Masses: clusters $+M$; voids $m = \delta_V \rho V < 0$ with $\delta_V = -0.70$.
- Pair sets: 3D Delaunay neighbours (primary), and as a check all pairs within 6 Gpc.
- Force statistics use interior bodies only (≤ 26 Gpc from us), to avoid the one-sided edge.

9.2 Results

- **Pair energies (foam s2, Delaunay pairs):** cluster–void repulsion $+4.7\times10^{69}$ J dominates cluster–cluster -2.1×10^{69} J and void–void -0.7×10^{69} J. The edits change each by 1–3%.
- **Moved bodies:** they end ~20% closer to force balance, and they are not pushed toward the ring: 43.5% of moved bodies have a net force toward the ring, against 50% for the rest.

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10. Compensated random web: move + resize under the entropy rule (completed)

10.1 Generator: distributions and references

How (`sequential_open/compweb_gen.py` → `final_comp_s<seed>.npz`, `compweb_s<seed>_gen.json`; shapes in `compweb_shapes.npz`).

- **Void sizes.** We use the *shape* of the project's void size function (`void_mass_network/lss.py`). This is the volume-conserving "Vdn" model of Jennings, Li & Hu 2013 (arXiv:1304.6087), built on the two-barrier excursion set of Sheth & van de Weygaert 2004 (astro-ph/0311260), with $\delta_v = -2.7$, $\delta_c = 1.686$ and $r = 1.7\, r_L$:

$$\frac{dn}{d\ln r} = \frac{f_{\rm SvdW}(\sigma(r_L))}{V(r)} \frac{d\ln\sigma^{-1}}{d\ln r_L}.$$

  It is sampled above 10 Mpc and scaled so that the median is 1.5 Gpc (the body scale of the foam webs). Range: 1.36–2.14 Gpc (5–95%).
- **Void depth.** Clampitt & Jain 2015 (arXiv:1404.1834) shallow voids: δ_v uniform in $[-0.45, -0.25]$, mean -0.35, as in `sequential/shallow/shallow_core.py`.
- **Cluster masses.** The *shape* of the Tinker et al. 2008 (arXiv:0803.2706) mass function for $\Delta = 200\times$ mean, with $A = 0.186$, $a = 1.47$, $b = 2.57$, $c = 1.19$:

$$\frac{dn}{d\ln M} = f(\sigma)\frac{\bar\rho_m}{M}\frac{d\ln\sigma^{-1}}{d\ln M}, \qquad f(\sigma) = A\left[\left(\tfrac{\sigma}{b}\right)^{-a} + 1\right] e^{-c/\sigma^2}.$$

  It is abundance-matched above $3\times10^{15}\, M_\odot$ ($M/M_{\rm min} \approx 1.0$–1.7) and scaled to a median of 1.40×10^{50} kg. The linear σ(R) is EH98, σ8 = 0.81, Ωm = 0.31, h = 0.70 (`density_vs_scale/lcdm_sigma_tophat.txt`).
- **Cluster size.** δ_c is log-uniform in 30–300 and $r_c = (3M_c/4\pi\delta_c\rho)^{1/3}$, giving a median of 0.155 Gpc.
- **Compensation.** As in §8.4, with the wall at R to $1.3R$, uniform in volume, random directions. The result is $N = 0.49 + M_{\rm def}/1.48\times10^{50}$ kg, about 4.5 clusters per void.
- **Placement.**
  - Void centres are uniform in volume, placed by random sequential addition, largest first, from a random draw of sizes.
  - A void is kept only if all of its clusters fit.
  - All the rules of §0.5 apply.
  - Filling stops at a void volume fraction of 30%, the same as the foam webs. The runs actually stopped at about 29%.

Web size. About 9,000 bodies per seed:

seed	voids	clusters	bodies
1	1,674	7,472	9,146
2	1,656	7,372	9,028
3	1,645	7,417	9,062

10.2 Fitter: unit edits that keep compensation exact

Plain words. To keep each void's balance intact, an edit never moves a void away from its own clusters and never changes a cluster's mass on its own. There are two kinds of edit "unit".

Formulas (`sequential_open/minedit_comp.py`, checkpoints `ckpt/minedit_comp_<tag>.pkl`).

- **Group unit** g: the void and all of its wall clusters together, with 4 parameters $(\boldsymbol{d}_g, a_g)$. Each member k goes to

$$\boldsymbol{x}_k = \boldsymbol{c}_g + \boldsymbol{d}_g + a_g^{1/3}\boldsymbol{u}_k, \qquad M_k = M_{k,0}\, a_g \frac{\rho(\boldsymbol{c}_g + \boldsymbol{d}_g)}{\rho(\boldsymbol{c}_g)}, \qquad r_k = r_{k,0}\, a_g^{1/3},$$

so the group's net mass stays exactly zero.

- **Cluster unit** k: one cluster slides inside its own wall, with 3 parameters $\boldsymbol{u}_k$, subject to $R_g + r_k + 0.02 \le |\boldsymbol{u}_k| \le 1.3R_g$ in the group frame. Its mass does not change.
- **Rules.** As in §0.5, plus the wall rows for cluster units, plus the entropy rule (§6.3).
- **Search.** Each step tries a fine-scale batch and, if that fails, a leverage batch.
  - The fine batch takes 8 near-ring units, ranked by their own m13+ power. Its starts are in place, pushed 2–8 Gpc away from the ring, and random within 15 Gpc.
  - If the fine batch is accepted, a 16-unit leverage "absorb" batch follows.
  - The leverage batch takes $\mathrm{clip}(n/6, 2, 24)$ units. Its gain is as in §6.1, with moves capped at 5 Gpc for groups and 2 Gpc for clusters.
  - Every 5 accepted steps, the last 30 units are refitted jointly.
- **Stops.** ≤ 1×, the plateau rule (< 1% gain over 300 units), 40 failures in a row, or a time cap.

10.3 Results (`minedit_comp/comp_s<seed>_entropy.json`; figures in `sequential_open/comp_web/` and `comp_web/s2/`; report `comp_web/REPORT.md`)

seed	start	end	bodies edited	units (group / cluster)	stop
3 (best)	789×	**0.998×**	1,099 (12.1%)	324 (191 / 133)	reached 1×
2	1,077×	1.18×	1,927 (21.3%)	1,046 (483 / 563)	plateau
1	1,260×	7.40×	2,538 (27.7%)	1,122	plateau

Seed 3 in detail

- **Moves and resizes:** 1,067 bodies moved (178 voids, 889 clusters) and 930 resized. The median move is 0.6 Gpc, 90% of moves are under 2.3 Gpc, and the largest is 6.5 Gpc. The median mass factor is 1.02.
- **Leftover by scale:** m1–12 goes 640× → 0.02×; m13+ goes 461× → 1.00×.
- **Symmetric share:** 67% → 35%.
- **Entropy, before → after** (none decreased):
  - radial −0.0456 → −0.0441
  - ring −0.0054 → −0.0050
  - angular −0.0110 → −0.0107
  - link_cc 3.0009 → 3.0037
  - link_cv 2.3176 → 2.3449
  - link_vv 2.0223 → 2.0666
  - ncount 2.4804 → 2.5002
- **Ring density:** bodies centred 0–3 Gpc from the ring went 151 → 160 in 2,530 Gpc³ (0.060 → 0.063 per Gpc³; the web averages 0.080). Bodies whose surface is within 3 Gpc went 217 → 216. No ring and no pile-up.
- **Compensation:** exact. The largest |group net mass| / deficit is 4×10^{-16}.

- **Slab mesh:**
  - triangle median 1.45 → 1.38 Gpc² over the whole slab
  - triangle median 1.84 → 1.59 Gpc² in the 0–3 Gpc band
  - connections per node: median 6 (mean 5.93)
  - median link 2.09 → 2.05 Gpc

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11. Caveats

- **Point masses.** Every body is a point mass for the kernel. Near the ring this exaggerates fine-scale (m13+) texture; it is the reason for the no-straddle rule and for the high starting leftovers of uniform webs.
- **Ring sampling.** 360 ring points (1°) alias structure finer than ~250 Mpc at the ring into the sampled modes. Treat m ≥ 13 values from the checkerboard and wave backgrounds as good to about a factor of 1.5.
- **Coarse-grained bodies.** The bodies are Gpc-scale stand-ins for whole regions of voids and clusters. The size and mass distributions keep only the *shapes* of the literature functions, rescaled to that body scale.
- **No dark matter, no light-cone evolution.** Everything is at z = 0, and we compare observables only. Baryon-only densities would scale the patterns down by about 6×.
- **The entropy rule is a choice.** Seven specific measures are used. A uniform random start is already near maximum disorder, so the rule rejects most moves there; the runs succeed, but slowly.
- **Energy estimates** (§5.3, §7.3) are order-of-magnitude or toy numbers. The faster-than-light edge needs a relativistic (LTB) treatment.
- **Sign convention.** With voids as negative mass, like-like pairs are bound and cluster–void pairs repel, which is the opposite of an ionic crystal. Without compensation this drives segregation.

Next steps raised

- Share walls between neighbouring voids in the balance rule.
- Test how results depend on the ring rule (2 or 5 Gpc, smooth penalty).
- Full 3D (tetrahedral) version of the mesh statistics.
- Relativistic LTB treatment for the faster-than-light edge.